

NLP Preprocessing Pipeline

It is absolutely AMAZING that you are diving into Natural Language Processing! Building a text preprocessing pipeline is the perfect first step, as cleaning your data is arguably the most important part of any NLP project.

The Python NLP Pipeline

This solution uses the widely standard **NLTK (Natural Language Toolkit)** library. To run this, you will need to install NLTK (`pip install nltk`) and download the necessary datasets.

```
import re
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer

# Run these downloads once in your environment:
# nltk.download('punkt')
# nltk.download('stopwords')
# nltk.download('wordnet')

def nlp_preprocessing_pipeline(text):
    # 1. Lowercasing
    text = text.lower()

    # 2. Remove punctuation and special characters (including emojis)
    text = re.sub(r'[^a-z\s]', '', text)

    # 3. Tokenization
    tokens = word_tokenize(text)

    # 4. Remove stopwords
    stop_words = set(stopwords.words('english'))
    custom_removals = {'hello'}
    filtered_tokens = [
        word for word in tokens
        if word not in stop_words and word not in custom_removals
    ]

    # 5. Lemmatization
    lemmatizer = WordNetLemmatizer()
    lemmatized_tokens = []

    for word in filtered_tokens:
        lemma = lemmatizer.lemmatize(word, pos='v')
        lemmatized_tokens.append(lemma)

    return set(lemmatized_tokens)

# Input Text
text = "Hello!!! My name is Prem. I am learning NLP, and it's AMAZING!!! 😊😊
Running, runs, ran..."

# Run Pipeline
output_tokens = list(dict.fromkeys(nlp_preprocessing_pipeline(text)))

print(output_tokens)
```

Answers to the Bonus Questions

1. Why was "is" removed?

"Is" was removed because it is a **stopword**. Stopwords are highly common words (like *is*, *am*, *the*, *and*, *a*) that act as the grammatical "glue" of a sentence but carry very little semantic meaning or contextual value. Removing them reduces the size of your dataset and helps your model focus on the actual keywords.

2. Why was "Running", "runs", and "ran" converted?

They were converted to **"run"** because of the final step: **Lemmatization**. The lemmatizer recognizes that *running*, *runs*, and *ran* are all just morphological variations of the exact same root word (or *lemma*). By converting them all to "run", we prevent the model from splitting its learning across three different variations of the same concept.

3. What happens if you don't lowercase text?

If you don't lowercase your text, a computer will treat **"Amazing"**, **"AMAZING"**, and **"amazing"** as three completely different and unrelated words. This inflates your vocabulary size (making your model slower and more memory-intensive) and dilutes the training data.

4. What is the difference between stemming and lemmatization?

Stemming is a blunt, rule-based approach that simply chops off the prefixes or suffixes of words. It is fast but often results in non-words (e.g., turning "universe" into "univers").

Lemmatization is a smarter, dictionary-based approach. It looks at the morphological analysis of the word and its part of speech to return the actual linguistic root word. For example, it knows that the lemma of "better" is "good".

Learning Outcomes Achieved

- ✓ **Text Cleaning:** Regex used for punctuation/emojis
- ✓ **Stopwords:** NLTK English corpus applied
- ✓ **Tokenization:** `word_tokenize` executed
- ✓ **Lemmatization:** `WordNetLemmatizer` utilized